

**Gao Haiyang** May 22, 2004

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## **EDUCATIONAL BACKGROUND**

### **∞ Wuhan University**

**09/2022-06/2026**

Degree: Bachelor of Finance & Bachelor of Mathematics

Major: Finance

Minor: Mathematics and Applied Mathematics

Mark: 3.65/4.0 (Major); 3.66/4.0 (Minor)

Core Modules: Monetary Banking (96), Corporate Finance (95), Options, Futures and other Derivatives (93), Methods of Data Analysis (91), Theory of Markets and Equilibrium (90), Real Analysis (95), Numerical Analysis (95), Stochastic Process (91), Machine Learning (90)

### **∞ Yale Summer Session**

**06/2025-08/2025**

Courses: Game Theory (course grade: A), YData: Introduction to Data Science (course grade: A)

### **∞ Graduate Record Examination**

General Test Scores: 322

## **PAPERS**

[2024 Green Finance Annual Report](#), 2025, Second author, Industry Publication

## **ACADEMIC PROJECT**

### **∞ Economic Effects of LA 2028 Olympics**

**08/2025-Present**

Position: Research Assistant, supervised by Dr. Lorena Martin, Trojans Sports Research Lab, USC

- Participating in a research project on analysing the effect of LA 2028 Olympics to local economy.

### **∞ Transformer-Based Modelling of Financial Time Series**

**07/2025-Present**

Position: Researcher, supervised by Dr. Shivam Sharma, Yale University

- Leading a research project on applying Transformer architectures to financial time series. Independently reviewed over 100 publications across relevant disciplines and presented findings to the supervisor for discussion and feasibility assessment.
- Developing refinements to the transformer-based pre-trained time-series model, including: 1) introducing sparsity to the model; 2) combining machine learning and statistical forecasting methods to improve model performance.
- Manuscript available [Here](#)

### **∞ Innovation and Pricing Research on Corporate Carbon Financial Products**

**03/2025-07/2025**

Position: Research Assistant

Sponsor: Research Group of Professor Ren Yujie, International Institute of Green Finance (IIGF), CUFE

- Participated in project Innovation and Pricing Research of Corporate Carbon Financial Products sponsored by IIGF, CUFE and State Grid Research Institute; attended weekly meetings and contributed quantitative finance expertise to address pricing challenges.

### **∞ Research on Platform Economy Theory**

**09/2024 – 07/2025**

Position: Research Assistant

Sponsor: Wuhan University, Research Group of Professor Shen Bo

- Responsible for daily screening of current situation of platform economy, collected information on development trends.
- Participated in exploring platform business models, market competition and user behaviour; discussed impact of platform economy on economic development.

### **∞ Quantitative Financial Studies**

**09/2024 - 07/2025**

Position: Research Assistant

Sponsor: China Southern Asset Management Co., Ltd., supervised by Dr. You Tao

- Studied quantitative finance, investigated different market factors to conduct yield analysis and forecasting.
- Conducted research on quantitative finance and current financial market; analysed data and composed reports to summarise study results.

### **Teaching Assistant for Game Theory Module in Wuhan University**

**02/2025-07/2025**

Position: Teaching Assistant

- Assisted instructor in Game Theory course, collected and graded assignments monthly, selected as an outstanding TA, earning high recognition from both the instructor and students.
- Provided in-depth explanations during exercise classes, including addressing students' questions on Cournot equilibrium, Bertrand equilibrium and other core models to strengthen their understanding of theoretical concepts.
- Organised Q & A sessions before quizzes and exams; supervised and graded mid-term and final exams, as well as collecting and grading monthly assignments.

### **INTERNSHIP**

#### **Industrial and Commercial Bank of China, Paid, Full-Time**

**08/2025-10/2025**

Position: Intern, Corporate Finance Department & International Business Department

- Served at the corporate finance department as an intern to learn the loan approval and customer credit evaluation process, analysed the existing market and financial risks.
- Analysed various financial instruments and documents used in trade financing services, such as letters of credit, import and export bills of lading, bills of exchange, and documentary collections, as well as evaluated their roles in mitigating risk and facilitating international trade transactions.

#### **China Securities Co., Ltd, Full-Time**

**07/2024-08/2024**

Position: Intern, Bond Underwriting Department

- Automated the processing of Wind database using Python, enabling the completion of weekly bond reports 40% with higher precision; published 5 weekly reports to provided interested parties with valid information for strategy adjustment.
- Supported statistical analysis for partnerships; completed automated data processing via Python and assisted in bond market data analysis with Wind to provide data support to team and help optimise business cooperation strategies.

#### **KPMG Huazhen LLP, Paid, Full-Time**

**01/2024-02/2024**

Position: Intern, Audit Department

- Participated in offline annual review of CRRC Corporation, experiencing entire audit process from determining audit objectives to composing audit report; developed Excel templates to automate data import and validation, improved team efficiency by 70%.
- Established communication channels between audit project team and client companies through inspecting audit objectives and collecting information; understood audit risks identification methods and audit risk models.

### **LANGUAGES**

Mandarin Chinese (native), English (proficient)

IELTS: Overall: 7.5 (Listening: 7.5, Reading: 9.0, Writing: 6.5, Speaking: 7.5)

### **PROGRAMMING SKILLS**

Python, The R Programming Language, STATA, LaTeX, MATLAB, Excel

### **HONOURS & AWARDS**

2<sup>nd</sup> class scholarship for academic year 2024-2025

3<sup>rd</sup> class scholarship for academic year 2023-2024

3<sup>rd</sup> Prize in National ETF Simulation Investment Elite Challenge

Won team championship 2 times as a key player, 2<sup>nd</sup> Place in Men's Doubles and 2<sup>nd</sup> Place in Mixed

Doubles, Hongyi Honours College Badminton Tournament